HW5 12209095 杜奕璇

1. $\chi^2 = \sum \frac{(O_i - E_i)^2}{E_i}$ $\because X \sim bin(3, 0.5)$ $n = 50$

$E_0 = \binom{3}{0}(0.5)^0 \times (0.5)^3 \times 50 = 6.25$

$E_1 = \binom{3}{1}(0.5)^1 \times (0.5)^2 \times 50 = 18.75$

$E_2 = \binom{3}{2}(0.5)^2 \times (0.5)^1 \times 50 = 18.75$

$E_3 = \binom{3}{3}(0.5)^3 \times 10.5)^0 \times 50 = 6.25$

$H_0: X \sim B(3, 0.5)$

$\sum \frac{(O-E)^2}{E} \sim \chi^2(3)$ $\chi^2 = \sum \frac{(O-E)^2}{E} = \frac{(6.25-15)^2}{6.25} + \frac{(18.75-20)^2}{18.75} + \frac{(18.75-10)^2}{18.75} + \frac{(6.25-5)^2}{6.25}$

$= \frac{50}{3} \approx 16.67 > \chi^2_{0.05}(3)$

$\chi^2_{0.05}(3) = 7.815$

$\therefore$ We will reject the H_0. $\therefore X$ does not distributes with $B(3, 0.5)$

(b) using MLE : $\hat{p} = \frac{\sum X}{nK} = \frac{(1\times20)+(2\times10)+(3\times5)}{50\times3} = \frac{55}{50\times3} = \frac{11}{30}$

(c) under $p = \frac{11}{30}$ $H_0: X \sim B(3, \frac{11}{30})$

X	0	1	2	3
frequency	15	20	10	5
Expectation	12.70	22.06	12.77	2.46

$\sum \frac{(O-E)^2}{E} \approx 3.83 < \chi^2_{0.05}(3) = 7.815$

$\therefore$ Do not reject H_0

$\therefore X$ distribute with $B(3, \frac{11}{30})$

2. $\hat{\lambda} = \bar{X} = \frac{53 + 2\times58 + 3\times39 + 4\times20 + 5\times5 + 6\times2 + 7\times1}{200} = 2.05$ $P(X=x) = \frac{(2.05)^x e^{-2.05}}{x!}$

$E(X=0) = \frac{(2.05)^0 e^{-2.05}}{0!} \times 200 \approx 25.75$

$E(X=1) = \frac{(2.05)^1 e^{-2.05}}{1!} \times 200 \approx 52.78$

$E(X=2) = \frac{(2.05)^2 e^{-2.05}}{2!} \times 200 \approx 54.10$

$E(X=3) = \frac{(2.05)^3 e^{-2.05}}{3!} \times 200 \approx 36.97$

$E(X=4) = 18.95$

$E(X=5) = 7.77$

$E(X=6) = 2.65$

$E(X=7) = 0.78$

$\sum \frac{(O-E)^2}{E} = 2.296$

P-value: $P[X \geq 2.296) = 0.89 > 0.05$

$\therefore$ do not reject H_0

$\therefore$ The distribution is Poisson(λ)

3. (a)

		30-50	50-60	60-70	70-80	80-100	total
I	A	4	4	3	2	2	15
II	B	1	3	3	4	4	15
		5	7	6	6	6	30

$\therefore$

E_{ij}	30-50	50-60	60-70	70-80	80-100
A	2.5	3.5	3	3	3
B					

$k=5 \quad r=2 \quad (k-1)(r-1)=4 \quad \chi^2 \sim \chi^2(4)$

$$\sum \frac{(E-O)^2}{E} = \frac{(4-2.5)^2}{2.5} + \frac{(1-2.5)^2}{2.5} + \frac{(4-3.5)^2}{2.5} + \frac{(3-3.5)^2}{3.5} + 2\times\frac{(3-3)^2}{3} + \frac{2\times(2-3)^2}{3} + 2\times\frac{(4-3)^2}{3}$$

$$= 3.276$$

$\chi^2_{0.05}(4) = 9.488 \quad \chi^2 < \chi^2_{0.05}(4) \quad \therefore$ Do not reject H_0.

$\therefore$ Class I and II have the same score distribution.

(b) $n=2,\ m=3$

$(n-1)(m-1)=2$

$\sum\frac{(O-E)^2}{E},\ \chi^2_{0.05}(2)=5.991$

	$X<60$	$60\le X<80$	$X\ge 80$	
I	8 (6)	5 (6)	2 (3)	15
II	4 (6)	7 (6)	4 (3)	15
	$\frac{12}{30}$	$\frac{12}{30}$	$\frac{6}{30}$	

H_0: Scores are independent between class I and II.

H_1: Scores are dependent between class I and II.

$\sum\frac{(O-E)^2}{E}=\frac{4}{6}+\frac{1}{6}+\frac{1}{3}+\frac{4}{6}+\frac{1}{6}+\frac{1}{3}=\frac{7}{3}<5.991$ ∴ Not reject H_0

∴ The scores and classes are independent

(c)

	$X<60$	$60\le X<80$	$X\ge 80$	
I	8	5	2	15
II	4	7	4	15
III	10	15	5	30
IV	20	30	20	70
	$\frac{42}{130}$	$\frac{57}{130}$	$\frac{31}{130}$	130

$(n-1)(m-1)=3\times 2=6$

$\chi^2_{0.05}(6)=12.59$

$\sum\frac{(O-E)^2}{E}\approx 5.242<12.59$ ∴ Not reject H_0

∴ The scores and classes are independent, the conclusion in (b) still hold.

4. H_0: $P_{win}=P_{spr}=P_{sum}=P_{fall}=\frac{1}{4}$

H_1: H_0 is not exist.

	winter	spring	summer	fall	
O	340	395	358	268	1361
E	$\frac{1361}{4}$	$\frac{1361}{4}$	$\frac{1361}{4}$	$\frac{1361}{4}$	

$\sum\frac{(O-E)^2}{E}=25.0779>\chi^2_{0.01}(3)$

$\chi^2_{0.01}(3)=11.34$

∴ Reject H_0 ∴ There is no relationship between weather conditions and the incidence of violent crime.

5. H_0: The working life of those four batches of bulbs is the same.
H_1: Have significant difference

	1	2	3	4	5	6	7	8
A	1600 (690)	1610 (737)	1650 (768)	1680 (812)	1700 (844)	1720 (1413)	1800 (985)	
B	1580 (1194)	1640 (1227)	1640 (1249)	1700 (1289)	1750 (1305)			
C	1460 (1911)	1550 (1918)	1600 (1998)	1640 (2548)	1660 (2685)	1740 (1597)	1820 (1124)	1820 (1124)
D	1510 (1424)	1520 (1464)	1530 (1487)	1570 (1526)	1600 (1554)	1680 (1190)		

$(4-1)\times(7-1)=3\times 7=21$ $\quad \chi^2_{0.05}(21)=32.67$

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ANOVA Test Statistic: 1.9327
p-value: 0.1538
Fail to reject the null hypothesis. The working life of the batches does not have a significant difference.
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6. Max: 335 Min: 305 $\frac{335-305}{10}=3$

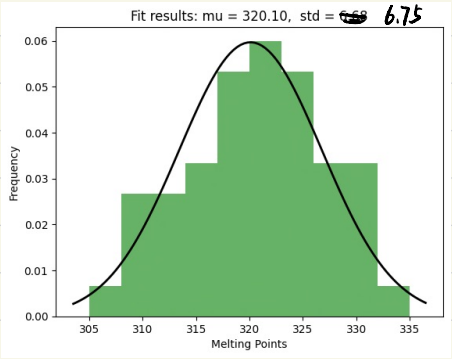
[332, 335]

[305, 308) [308, 311) [311, 314) [314, 317) [317, 320) [320, 323) [323, 326) [326, 329) [329, 332)

0 1 4 4 5 8 9 8 5 5 1

$\bar{x}=320.1$, $s^2=45.5612$

$Z=\frac{X-320.1}{\sqrt{45.5612}}\sim N(0,1)$




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Statistics=0.989, p-value=0.919
Fail to reject the null hypothesis. The data appears to be normally distributed.
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7.

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Results for (a):
          Categories  Observed     Expected
0           < -0.001      3949  4200.964476
1  [-0.001, -0.0004)       122    94.803029
2       [-0.0004, 0)        47    63.232494
3        [0, 0.0004)       277    63.232494
4    [0.0004, 0.001)       117    94.803029
5           >= 0.001      4205  4200.964476

Chi-square test statistic: 754.9576
Critical value at alpha=0.05 with 4 degrees of freedom: 9.4877
Do we reject the null hypothesis? Yes

Results for (b):
          Categories  Observed     Expected
0           < -0.001      3949  4055.878893
1  [-0.001, -0.0004)       122    96.248951
2       [-0.0004, 0)        47    64.258910
3        [0, 0.0004)       277    64.307686
4    [0.0004, 0.001)       117    96.504886
5           >= 0.001      4205  4340.800673

Estimated mu: 0.0009, Estimated sigma: 0.0216
Chi-square test statistic: 726.4047
Critical value at alpha=0.05 with 4 degrees of freedom: 9.4877
Do we reject the null hypothesis? Yes
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